Fengtong (Farah) Du

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ABOUT ME

Machine learning researcher with training in computer vision and a PhD background in computational neuroscience. I began my career developing **object detection, image and video analysis**, and **generative models**, and later extended these skills to modeling neural responses in visual cortex. My research centers on **representation learning, deep neural encoding models, and unsupervised learning** of visual features, combining large-scale neural recordings with modern vision architectures, developing PyTorch-based vision models, and studying how biological and artificial systems learn invariant, generalizable features.

EDUCATION

PhD in Neuroscience, Johns Hopkins University & HHMI Janelia **2021.8-present**

- Supervised by Carsen Stringer.
- Research focus: deep learning–based neural encoding models, unsupervised representation learning, computational modeling of visual cortex.
- Publications: *Nature Communications (2025)*, *Nature Neuroscience (2025).*

MS in New Artificial Intelligence and Media Technology, CUC **2017.9-2021.6**

- GPA: 3.9/4.0. Rank 1st

RESEARCH AND CAREER EXPERIENCE

PhD Researcher (Joint Program with HHMI Janelia) **2022.6–present**

Johns Hopkins University & HHMI Janelia

Project A: Unsupervised Learning of Visual Representations in Mouse Cortex (ongoing)

- Developed modeling framework for **unsupervised category learning**, quantifying how neural representations evolve with prolonged visual exposure without task supervision.
- Analyzed large-scale neural time series across V1 and medial areas to evaluate **hierarchical representation formation** and invariance.
- Modeling representational plasticity with **contrastive / temporal learning–inspired rules**, connecting learning dynamics with neural manifold geometry.
- Identifying sequence-dependent representational structure consistent with **temporal contrastive learning**, revealing sensitivity to transitions rather than isolated frames.

Project B: Deep Neural Encoding Models of Visual Cortex (Nature Communications 2025)

- Built a biologically inspired **two-layer single-neuron "minimodel"** in PyTorch predicting cortical responses to natural images.
- Designed cluster-based workflows to run thousands of independent models in parallel across **multiple GPUs**.
- Achieved high accuracy across mouse V1 and higher areas; aligned deep features with neuronal tuning properties.
- Work bridges sensory neuroscience with core ML representation learning.

Project C: Population Representation Analysis (Nature Neuroscience 2025)

- Contributed to **Rastermap**, a high-dimensional representation discovery method for large-scale neural recordings.
- Implemented scalable visualization and manifold learning tools enabling unsupervised pattern discovery in neural populations.

Machine Learning Reaserch Intern **2020.7-2020.10**

Fujitsu Research and Development Center

- Developed **GAN-based few-shot domain adaptation** and dataset expansion framework.

Computer Vision Intern **2017.2-2017.8**

Beijing Institute of Collaborative Innovation

- Built Faster-RCNN systems for tiny **object detection**, applied to sports video analytics and medical imaging.
- Developed end-to-end C++/Python pipelines for video detection tasks.

SELECTED PROJECTS

Texture Recognition in Neural Populations (COSYNE 2021)

- Compared CNN-based and cortical representations for **invariant texture recognition**.
- Demonstrated emergence of unsupervised invariance in biological systems.

Audio-Visual Integration Model (IJCNN 2021)

- Designed biologically motivated **continual learning** model using HH equation and STC learning rules.
- Proposed non-backprop learning strategies for robust **multimodal feature learning.**

PUBLICATIONS

1. **Fengtong Du***, M Ángel Núñez-Ochoa, M Pachitariu*, C Stringer*, A simplified minimodel of visual cortical neurons, Nature Communications, 2025, [link]
2. C Stringer, L Zhong, A Syeda, **Fengtong Du**, M Kesa, M Pachitariu*, Rastermap: a discovery method for neural population recordings, Nature Neuroscience, 2025, [link]
3. L Chen, S Zhu, W Chen, L Min, Y Zhao, **Fengtong Du**, S Guo, J Xing, Z Zhang, M Ji*, Gastroenterologist-level detection of gastric precursor lesions and neoplasia with a deep convolutional neural network, MedRobot, 2023, [link]
4. **Fengtong Du**, Sonia Joseph, Marius Pachitariu*, Carsen Stringer*, *Invariant texture recognition in mouse visual cortex*, submitted, cosyne 2021, [link]
5. Lihong Cao*, **Fengtong Du**, Wenjie Chen, *A neural assembly learning method for concept-like cells formation and continual learning*, *Granted Invention Patent*, National Intellectual Property Administration of China (CNIPA), Patent No. ZL202110427115.9, 2021.
6. Wenjie Chen, **Fengtong Du**, Ye Wang, Lihong Cao*, A *biological plausible audio-visual integration model for continual lifelong learning*, IJCNN, 2021, [link]
7. Wenjie Chen, **Fengtong Du**, Ye Wang, Lihong Cao*, *Predropout&Inhibition: A Brain-Inspired Method for Convolutional Neural Network*, CISP, 2018, [link]
8. **Fengtong Du***, Yanzhuo Zhou, Wenjie Chen, Lei Yang, *Bullet hole detection using series Faster-RCNN and video analysis*, ICMV, 2019, [link]
9. Xinyi Zhou*, Wei Gong, WenLong Fu, **Fengtong Du**, Application of deep learning in object detection, ICIS, 2017, [link]

SKILLS

Machine Learning: PyTorch, TensorFlow, representation learning, self-supervised learning, contrastive learning, generative modeling, CNNs, encoding models, dimensionality reduction, large-scale training pipelines
Programming: Python (expert), PyTorch, TensorFlow, NumPy/SciPy, MATLAB, C/C++
Data: large-scale neural datasets, two-photon imaging, multimodal sensory data, video & image data
Analysis: representation geometry, correlation analysis, PCA, unsupervised clustering, neural decoding
Other: version control (Git), scientific communication, mentoring, teaching